

Additional Heaps Problems

Filename: No File Submission

Instructions

This worksheet contains additional practice problems on heaps. Try to complete each problem on your own before looking at the solutions. The solutions will be provided separately, so make sure to give each problem an honest attempt first. Checking the answer too early defeats the purpose of the practice. As you work, time yourself and do not spend more than 5 minutes on any single problem.

For the following problems, assume that the root of the heap is stored at index 0. Furthermore, assume that the following helper functions are already implemented. These functions are defensive, meaning they safely handle invalid indices and perform any necessary bounds checking internally.

```
void percolateUp(int arr[], int arrSize, int upIndex);  
// Performs a percolate-up operation starting at index upIndex.  
// Note: arrSize refers to the number of items (i.e., heapSize).  
  
void percolateDown(int arr[], int arrSize, int downIndex);  
// Performs a percolate-down operation starting at index downIndex.  
// Note: arrSize refers to the number of items (i.e., heapSize).
```

Important Note

Percolate-up corresponds to the operation referred to in class as heapify-up, while percolate-down corresponds to heapify-down.

TODO 1

Complete the function `heapDecrease()` that decreases the key of an element at a specified index in a min-heap. The function should accept the following parameters: the heap array, the current number of elements in the heap, the index of the element whose key is to be decreased, and the new value.

If the given index is invalid, or if the new value is greater than or equal to the current value at that index, the function should return 0. Otherwise, update the value, restore the min-heap property, and return 1.

```
int heapDecrease(int heapArr[], int heapSize, int idx, int key) {
```

```
}
```

TODO 2

Complete the function that doubles the value of the minimum element in a min-heap and then restores the min-heap property. The function should take as parameters the heap array and the current number of elements in the heap.

```
void doubleMin(int heapArr[], int heapSize) {
```

```
}
```